

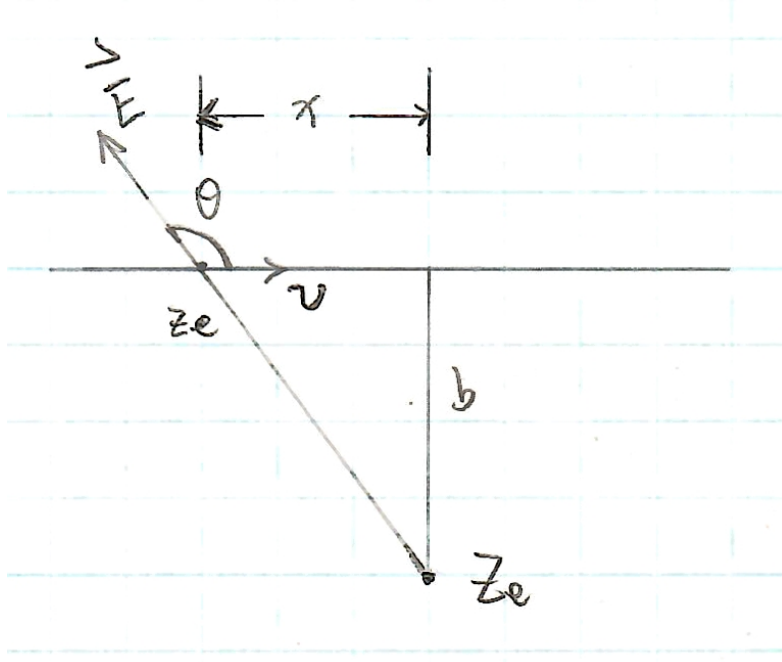


Figure 1: Relativistic particle moving in a straight line.

14.8 Solution: From Eq. (12.1), the equation of motion for a particle with mass m and charge ze in an electric field is

$$\frac{d}{dt}(\gamma mc\boldsymbol{\beta}) = \gamma^3 mc\boldsymbol{\beta}(\boldsymbol{\beta} \cdot \dot{\boldsymbol{\beta}}) + \gamma mc\dot{\boldsymbol{\beta}} = ze\mathbf{E}.$$

Dot by $\dot{\boldsymbol{\beta}}$ on both sides, we will have

$$\gamma^3 mc\beta^2(\boldsymbol{\beta} \cdot \dot{\boldsymbol{\beta}}) + \gamma mc(\boldsymbol{\beta} \cdot \dot{\boldsymbol{\beta}}) = \gamma mc(\gamma^2\beta^2 + 1)(\boldsymbol{\beta} \cdot \dot{\boldsymbol{\beta}}) = \gamma^3 mc(\boldsymbol{\beta} \cdot \dot{\boldsymbol{\beta}}) = ze\boldsymbol{\beta} \cdot \mathbf{E}.$$

Therefore,

$$\boldsymbol{\beta} \cdot \dot{\boldsymbol{\beta}} = \frac{ze}{\gamma^3 mc}\boldsymbol{\beta} \cdot \mathbf{E},$$

and

$$\dot{\boldsymbol{\beta}} = \frac{1}{\gamma mc} \left(ze\mathbf{E} - \gamma^3 mc\boldsymbol{\beta}(\boldsymbol{\beta} \cdot \dot{\boldsymbol{\beta}}) \right) = \frac{ze}{\gamma mc} (\mathbf{E} - \boldsymbol{\beta}(\boldsymbol{\beta} \cdot \mathbf{E})).$$

Using Lamor's formula, Eq. (14.26), the radiated power is

$$P(t) = \frac{2}{3} \frac{z^2 e^2}{c} \gamma^6 \left[|\dot{\boldsymbol{\beta}}|^2 - |\boldsymbol{\beta} \times \dot{\boldsymbol{\beta}}|^2 \right].$$

Since

$$|\dot{\boldsymbol{\beta}}|^2 = \frac{z^2 e^2}{\gamma^2 m^2 c^2} \left[|\mathbf{E}|^2 - 2|\boldsymbol{\beta} \cdot \mathbf{E}|^2 + \beta^2 |\boldsymbol{\beta} \cdot \mathbf{E}|^2 \right],$$

and

$$|\boldsymbol{\beta} \times \dot{\boldsymbol{\beta}}|^2 = \frac{z^2 e^2}{\gamma^2 m^2 c^2} |\boldsymbol{\beta} \times \mathbf{E}|^2 = \frac{z^2 e^2}{\gamma^2 m^2 c^2} \left[\beta^2 |\mathbf{E}|^2 - |\boldsymbol{\beta} \cdot \mathbf{E}|^2 \right],$$

then

$$\begin{aligned}
P(t) &= \frac{2}{3} \frac{z^4 e^4}{m^2 c^3} \gamma^4 \left[|\mathbf{E}|^2 - 2 |\boldsymbol{\beta} \cdot \mathbf{E}|^2 + \beta^2 |\boldsymbol{\beta} \cdot \mathbf{E}|^2 - \beta^2 |\mathbf{E}|^2 + |\boldsymbol{\beta} \cdot \mathbf{E}|^2 \right] \\
&= \frac{2}{3} \frac{z^4 e^4}{m^2 c^3} \gamma^4 (1 - \beta^2) \left[|\mathbf{E}|^2 - |\boldsymbol{\beta} \cdot \mathbf{E}|^2 \right] \\
&= \frac{2}{3} \frac{z^4 e^4}{m^2 c^3} \gamma^2 \left[|\mathbf{E}|^2 - |\boldsymbol{\beta} \cdot \mathbf{E}|^2 \right].
\end{aligned}$$

This can also be obtained from Prob. 14.11 (a), with $\mathbf{B} = 0$.

Assuming straight path of the particle, we have

$$|\mathbf{E}| = \frac{Ze}{x^2 + b^2}, \quad \mathbf{E} \cdot \boldsymbol{\beta} = -\frac{Ze}{x^2 + b^2} \frac{\beta x}{\sqrt{x^2 + b^2}},$$

and

$$P(t) = \frac{2}{3} \frac{z^4 Z^2 e^6}{m^2 c^3} \gamma^2 \frac{1}{(x^2 + b^2)^2} \left(1 - \frac{\beta^2 x^2}{x^2 + b^2} \right).$$

Integrating the power, and notice that

$$dt = \frac{dx}{v} = \frac{dx}{c\beta},$$

the total energy radiate is then

$$\begin{aligned}
\Delta W &= \int_{-\infty}^{+\infty} P(t) dt = \frac{2}{3} \frac{z^4 Z^2 e^6}{m^2 c^4 \beta} \gamma^2 \int_{-\infty}^{+\infty} \left[\frac{1}{(x^2 + b^2)^2} - \frac{\beta^2 x^2}{(x^2 + b^2)^3} \right] dx \\
&= \frac{2}{3} \frac{z^4 Z^2 e^6}{m^2 c^4 \beta} \gamma^2 \left[\frac{\pi}{2b^3} - \frac{\pi\beta^2}{8b^3} \right] = \frac{2}{3} \frac{\pi z^4 Z^2 e^6}{m^2 c^4 \beta} \gamma^2 \left[\frac{3}{8} + \frac{1 - \beta^2}{8} \right] \frac{1}{b^3} \\
&= \frac{2}{3} \frac{\pi z^4 Z^2 e^6}{m^2 c^4 \beta} \gamma^2 \left[\frac{3}{8} + \frac{1}{8\gamma^2} \right] \frac{1}{b^3} = \frac{\pi z^4 Z^2 e^6}{4m^2 c^4 \beta} \left(\gamma^2 + \frac{1}{3} \right) \frac{1}{b^3}.
\end{aligned}$$

For non-relativistic motion, $v_0 = c\beta$ and $\gamma \rightarrow 1$, this reduces to the result of Prob. 14.7.